

# Sourav Saha

SENIOR MACHINE LEARNING ENGINEER | GENERATIVE AI | COMPUTER VISION | NLP

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“Be the change that you want to see in the world.”

## Summary

Senior Machine Learning Engineer with 4+ years of experience designing, training, and deploying production-ready AI systems across LLMs, Computer Vision, ASR/TTS, and Generative AI. Proven ability to deliver end-to-end ML solutions — from data collection and model training to scalable API deployment. Experienced in YOLO-based CV systems, LLM-powered chatbots, RAG pipelines, anti-spoofing, and multimodal learning, with published research in vision-language modeling.

## Education

### United International University

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING (BSCSE)

Dhaka, Bangladesh

Mar. 2015 - Sept. 2021

### Jamuna Sar Karkhana Uchcha Madhyamic Bidhyalaya

HIGHER SECONDARY CERTIFICATE (HSC)

Jamalpur, Bangladesh

Aug. 2014

### Jamuna Sar Karkhana High School

SECONDARY SCHOOL CERTIFICATE (SSC)

Jamalpur, Bangladesh

May. 2012

## Work Experience

### Technohaven Company Ltd.

SR. SOFTWARE ENGINEER AND AI TEAM LEAD

Dhaka, Bangladesh

Nov. 2023 - Present

- Built an end-to-end pipeline that OCRs scanned Bangladeshi building plan documents using PaddleOCR and an LLM to extract structured data, then matches them against official government records via 7 custom algorithms (fuzzy name, property number, geo-weighted scoring). Includes a real-time multi-user review web app built with FastAPI and vanilla JS, where reviewers confirm or reject matches side-by-side with live image preview.
- Deployed a quantized YOLO-based NID detection model locally, processing 50K images per day with p95 latency under 150 ms; implemented CloudWatch monitoring and an automated retraining pipeline using MLflow.
- Architected and deployed a LangChain-based chatbot with PostgreSQL session persistence and Redis caching, reducing latency by 4%, and enabling WebSocket real-time streaming; validated through load testing to support 500 concurrent users with 99.9% uptime.
- Architected an end-to-end financial document processing pipeline leveraging PaddleOCR for table extraction with 95% accuracy, complemented by a human-in-the-loop correction interface to address the remaining 5%; enabled systematic data storage and continuous learning to retrain PaddleOCR and Qwen LLM models.
- Led the end-to-end development of an AI-powered Virtual Try-On platform, combining production-grade computer vision pipelines with VITON-based generative models, fine-tuned for traditional garments such as Panjabi and sari to deliver realistic, user-ready visual outputs.
- Led the architecture and implementation of a real-time IELTS Speaking simulator, integrating streaming ASR, LLM-based rubric-aligned scoring, and session orchestration to emulate official exam conditions.
- Led a team in developing and training Text-to-Speech (TTS) and Automatic Speech Recognition (ASR) models from scratch for Bangla and English languages.

### Anchorblock Technology Ltd.

MACHINE LEARNING ENGINEER

Dhaka, Bangladesh

Mar. 2022 - Oct. 2023

- Developed an LLM-based community management bot for real-time spam message classification, improving moderation efficiency.
- Built an AI chatbot using LangChain and LLaMA-2, enhancing automated conversational support for organizational use cases.
- Implemented AI-driven e-commerce features, including virtual try-on, image in-painting, and text-to-image design generation.
- Applied factor analysis and machine learning models to design data-driven stock market trading strategies.

## Projects

### kolvax

BUILDING A NO-CODE AI VOICE AUTOMATION PLATFORM WHERE USERS CAN CREATE WORKFLOWS, ASSIGN PHONE NUMBERS, AND DEPLOY VOICE BOTS TO HANDLE CALLS, COLLECT INFORMATION, AND EXECUTE BUSINESS PROCESSES AUTONOMOUSLY.

Twilio, TTS, ASR, Pipecat

Apr. 2026

## Publication

- |      |                                                                                                                                                                                                                              |                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2026 | <b>A Controller-Mediated Multi-Agent LLM Framework for Real-Time Digital Banking Workflow Execution,</b><br>Arafat Md Easin; Saha Sourav; Orosz Tamás                                                                        | (Under Review) |
| 2024 | <b>An Intelligent LLM-Powered Personalized Assistant for Digital Banking Using LangGraph and Chain of Thoughts,</b> Arafat Md Easin; Saha Sourav; Orosz Tamás                                                                |                |
| 2023 | <b>BSTE-GAN: Generative Adversarial Network guided Bengali Scene Text Editing System,</b> Sourav Saha, Md. Easin Arafat Md Aminul Haque Palash, Faria Afrin, Dewan Md Farid and M. Shamim Kaiser                             | (Under Review) |
| 2023 | <b>Empowering Real-Time Insights through LLM, LangChain, and SAP HANA Integration,</b> Md. Easin Arafat, Sourav Saha, Asuah Georgina, Tamas Orosz                                                                            |                |
| 2023 | <b>Simulated data for bangali text localisation in natural image,</b> Sourav Saha, Md Easin Arafat, Md Aminul Haque Palash, Dewan Md Farid, M Shamim Kaiser                                                                  |                |
| 2021 | <b>Bangla Image Caption Generation through CNN-Transformer based Encoder-Decoder Network,</b> Md Aminul Haque Palash, MD Abdullah Al Nasim, Sourav Saha, Faria Afrin, Raisa Mallik and Sathishkumar Samiappan                |                |
| 2021 | <b>Covid-19 in bangladesh: an exploratory data analysis and prediction of neurological syndrome using machine learning algorithms based on comorbidity,</b> Aditi sarkar, Sourav Saha, Shuvo chandra Das, Partha Chakraborty |                |

## Certifications

### Generative AI with Large Language Models

DEEPLARNING.AI

Jul. 2023

### LangChain for LLM Application Development

DEEPLARNING.AI

May 2023

### Building and Evaluating Advanced RAG Applications

DEEPLARNING.AI

Mar 2021

## Skills

|                             |                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Programming Language</b> | Python, C, C++, Java, Javascript                                                                        |
| <b>Database</b>             | SQL, PostgreSQL, MongoDB                                                                                |
| <b>ML Tools</b>             | Scikit-learn, Matplotlib, keras, Tensorflow, Pytorch, Pandas, NumPy, Zipline, Pyfolio , LangChain, Yolo |
| <b>Caching</b>              | Redis                                                                                                   |
| <b>Web Scraping</b>         | Beautifulsoup, Selenium                                                                                 |
| <b>Web Framework</b>        | Django, Flask, FastAPI                                                                                  |
| <b>Version control</b>      | Github                                                                                                  |
| <b>Familiar Language</b>    | Pyspark, Docker, hadoop                                                                                 |
|                             | English, Bangla                                                                                         |